

Online Distribution Shaping Report

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Online Distribution Shaping for Policy Learning

Executive assessment

The literature treats training-distribution shaping as several related but distinct problems. A selector may reweight a large data pool, move a task distribution toward a target through optimal transport, generate adversarial environments, curate a replay buffer, or allocate physical state-action samples under flow constraints. These mechanisms all change what the downstream policy sees, but they optimize different objectives and support different kinds of theory.

The central conclusion is a theory gap. Bilevel data-selection methods can improve downstream performance but generally lack regret or convergence guarantees in their neural, online form [Gra23]. Optimal-transport curricula provide principled geometry and stage-to-stage transfer bounds, but not cumulative online regret for the shaping policy [Hua22b, Kli22b, Kli23]. Unsupervised environment-design methods provide minimax-regret or equilibrium results, but these are conditional on equilibrium and usually do not scale with the complexity of a restricted distribution-policy class [Den20, Jia21, Par22]. The strongest finite-sample results come from structured exploration and data-acquisition settings, whose objectives are reachability, policy identification, or sampling efficiency rather than aggregate downstream performance [Tar21, Hu26].

A useful target for future work is therefore a finite-time, high-probability regret bound for a restricted family of distribution-shaping policies, where feedback is the downstream performance of the policy being trained.

A common problem formulation

Let x denote a training context, task, environment parameter, or data item. At round t , a shaping policy selects a distribution q_t from a restricted class

$$q_t = q_{\phi_t}, \quad \phi_t \in \Phi.$$

The downstream learner updates a policy π_t using data drawn from q_t . Performance is evaluated under a target distribution μ :

$$J(\pi_t; \mu) = \mathbb{E}_{x \sim \mu}[V^{\pi_t}(x)].$$

A natural comparator is the best shaping policy in Φ . A corresponding regret target would compare the cumulative downstream performance obtained by the online selector with the performance obtained by the best fixed or slowly changing shaping policy in the class. This is not the objective analyzed by most existing papers. Many instead optimize one of the following:

- minimax regret against an adversarial environment;
- convergence toward a target task distribution;
- the value gap between successive curriculum stages;

- the probability of discovering reachable goals;
- the probability of selecting the optimal policy;
- the generalization gap induced by training-level selection.

Keeping these objectives separate is essential. The word “regret” is used both for a learner-versus-teacher performance gap and for cumulative online regret, but the two are not interchangeable.

Main families of methods

Family	Shaped object	Typical restriction	Main guarantee	Main limitation
Bilevel data selection	Weights over a generic data pool	Neural weighting network with normalized positive weights	Empirical downstream gains	No formal regret or convergence theorem for the online neural method [Gra23]
Self-paced and OT curricula	Context or task distribution	Gaussian, particle, or trust-region distributions	Local-optimum, transfer-gap, or target-convergence results [Kli21, Hua22b, Kli22b]	No cumulative regret for the distribution selector
Adversarial and replay curricula	Environment or level distribution	Teacher policies, replay buffers, or evolutionary curation	Minimax-regret or equilibrium results	Guarantees are conditional on reaching equilibrium [Den20, Jia21, Par22]
Goal-directed exploration	Goal or state-selection distribution	Reachability and uncertainty constraints	PAC and near-minimax exploration bounds [Tar21]	Optimizes discovery and reachability, not general target-distribution performance
Online data acquisition	State-action sampling allocation	Flow-balance and persistent-exploration constraints	Large-deviation and near-robust optimality results [Hu26]	Optimizes policy identification rather than policy-training performance
Robust self-paced curricula	Robustness-budget distribution	Scalar or narrow parametric curriculum	Finite-time stationarity and curriculum-gap bounds [Sat25b]	The curriculum variable is highly specialized

Bilevel optimization of training distributions

Adaptive Training Distributions formulates data selection as a bilevel problem. A weighting network assigns normalized positive weights to examples in a generic dataset, while an outer objective measures performance on a more specific dataset [Gra23]. The online algorithm alternates updates to the main model and the weighting network, using differentiable selection, stochastic bilevel optimization, or gradient-alignment updates.

This is the closest direct match to general training-data distribution shaping. The restriction is explicit: the selector can only produce distributions representable by the weighting network. The method also highlights a practical issue that any regret analysis must address: the downstream objective depends on the result of an incompletely solved, nonconvex inner training problem.

The paper reports useful empirical gains in language modeling and translation, but does not prove regret, convergence, or sample-complexity guarantees for the full neural procedure. The authors explicitly point to nonconvexity and nonunique inner solutions as obstacles to applying standard bilevel theory [Gra23]. Thus, the method supplies a strong algorithmic template but not the desired restricted-class guarantee.

Self-paced and optimal-transport curricula

Self-paced reinforcement learning makes the training distribution itself an optimization variable. The learner balances current performance against proximity to a target context distribution and constrains each update to remain close to the previous distribution [Kli21]. In the practical algorithms, the context distribution is primarily Gaussian, so the restricted policy class is a low-dimensional parametric family.

The theory establishes a majorize-minimize interpretation and connects alternating updates to local optimization, but it does not give online regret for choosing among context distributions. The Gaussian restriction is also important: it makes the KL constraints tractable while limiting multimodal or nonparametric curricula [Kli21].

Optimal transport broadens the allowable distribution family and adds geometry to the update. GRADIENT constructs a Wasserstein path between a source and target distribution. Under assumptions on contextual MDPs, the value gap between successive optimal policies is bounded by the Wasserstein distance between successive curriculum distributions:

$$V^{\pi_{k+1}^*}(\rho_{k+1}) - V^{\pi_k^*}(\rho_{k+1}) \leq mW_{d^*}(\rho_k, \rho_{k+1}).$$

For constant-speed interpolation, the gap scales with the interpolation step and the Wasserstein distance between source and target [Hua22b]. This is a useful transfer guarantee: small distributional moves make successive stages easier to inherit. It is not a regret bound because it compares adjacent optimal policies rather than cumulative performance of an online selector.

CURROT instead solves a constrained transport problem that moves a particle-based curriculum toward a target distribution while retaining only contexts on which the current policy exceeds a performance threshold. A Wasserstein trust region limits movement from the previous curriculum [Kli22b, Kli23]. This gives three forms of restriction:

- the distribution is represented by finitely many particles;
- updates remain within a Wasserstein radius of the previous distribution;
- support is restricted to contexts that meet a competence threshold.

The papers show strong empirical performance on multimodal, nonuniform, and partly infeasible task spaces. Their formal claims concern convergence toward the target under a strong-learner assumption and the benefits of metric-aware interpolation. CURROT does not provide a cumulative online regret guarantee [Kli22b, Kli23].

The OT contribution is therefore best understood as a geometry and constraint mechanism. It can make the class $\{q_\phi : \phi \in \Phi\}$ richer than a Gaussian family while controlling update size, but it does not by itself specify the statistical objective or prove that the selected transport path is regret-optimal.

Adversarial and replay-based distribution shaping

PAIRED frames environment design as a game. An adversary chooses environment parameters, an antagonist tries to solve the resulting environment, and a protagonist is trained to close the antagonist’s performance gap. The teacher’s objective is a regret-like difference between antagonist and protagonist returns [Den20]. At Nash equilibrium, the protagonist is a minimax-regret policy under the paper’s assumptions.

This is a genuine regret formulation, but it is not the standard online-learning comparator of the best distribution policy in a restricted class. The guarantee is a game-theoretic statement about equilibrium behavior. It also depends on the teacher and learner having sufficient capacity and on the optimization process reaching an equilibrium.

REPAIRED combines an active generator with a replay curator. The theoretical analysis shows that equilibria of the dual-curriculum game inherit approximate or minimax-regret properties under the stated assumptions [Jia21]. ACCEL replaces the learned generator with random search and evolutionary edits while retaining regret-based curation; its robustness claim is likewise inherited from the replay framework and is conditional on reaching equilibrium [Par22].

These methods make an important conceptual point: shaping can mean restricting training to a small, evolving subset of environments rather than sampling broadly. But the restriction is defined by estimated regret or replay scores, not by a prespecified parametric class whose statistical complexity enters a finite-time bound.

Structured exploration and data acquisition

The closest finite-sample theories arise when distribution shaping is viewed as selecting goals or state-action samples for exploration.

AdaGoal chooses goals that are estimated to be reachable but uncertain. In tabular MDPs, the method has a nearly minimax sample-complexity guarantee of

$$\tilde{O}(L^3 S A \epsilon^{-2}),$$

under a reset-action and reward-free exploration setup. A linear-mixture extension gives a structured function-approximation result [Tar21]. The algorithm shapes the goal distribution toward the current reachability frontier, but its objective is to discover and master reachable goals rather than optimize expected downstream performance under an exogenous target distribution.

Optimal Data Acquisition studies an even more explicit restricted-distribution problem. It optimizes a stationary state-action allocation ω , but feasible allocations must obey the MDP’s flow-balance constraints. The resulting class is not arbitrary probability distributions; it is the set of occupancies physically realizable by the dynamics. A lazy projected subgradient method updates the allocation online, and the paper proves almost-sure convergence to a surrogate optimum together with near-robust optimality for policy identification [Hu26].

This work is a useful model for how a future distribution-shaping theory might handle restrictions. The feasible class is explicit, the update is online, and the guarantee reflects the geometry of the constraint set. What remains different is the objective: the method optimizes the exponential decay rate of incorrect policy selection, not the downstream return achieved by a policy trained on the selected data.

Robustness and generalization objectives

Some recent methods shape distributions to control robustness or generalization rather than regret.

Distributionally Robust Self-Paced Curriculum Reinforcement Learning gradually increases a robustness budget. Its curriculum distribution is effectively a point mass over the current budget, with trust-region and budget constraints. The paper proves finite-time stationarity and a bound on the gap between performance at the current curriculum and at the target robustness budget [Sat25b]. This is closer to an optimization convergence theorem than an online regret result, but it demonstrates how a restricted curriculum class can enter a finite-time analysis.

Data-Regularised Environment Design uses a generative model to keep generated levels near the observed training distribution and adaptively samples from a replay buffer. Its main theory bounds the generalization gap through mutual information between training levels and the learned policy [Gar24]. This addresses distribution shift and overfitting, not cumulative regret. It is nevertheless relevant because a distribution shaper can improve training performance while harming target-distribution performance if its generated data leave the intended support.

Comparison of guarantee types

Guarantee type	Representative papers	What is controlled	Why it is not yet the desired result
Minimax regret at equilibrium	[Den20, Jia21, Par22]	Worst-case gap between protagonist and antagonist or optimal teacher	Conditional on equilibrium; no complexity-sensitive finite-time bound
Stage-to-stage transfer gap	[Hua22b]	Loss from transferring a policy between nearby distributions	Does not evaluate cumulative online selection
Target-distribution convergence	[Kli21, Kli22b, Kli23]	Movement of the curriculum toward a target under competence constraints	No bound against the best restricted shaper
PAC exploration	[Tar21]	Discovery of reachable or uncertain goals	Different objective and stronger access assumptions
Allocation optimality	[Hu26]	Efficiency of physically feasible state-action sampling	Optimizes policy identification, not downstream training return

Guarantee type	Representative papers	What is controlled	Why it is not yet the desired result
Stationarity and curriculum gap	[Sat25b]	Convergence of policy and robustness-budget iterates	Specialized scalar curriculum and no comparator regret
Generalization gap	[Gar24]	Performance loss from dependence on selected training levels	Does not measure online decision quality

What a restricted-class regret theorem would need

A useful theorem would have to connect four layers that existing papers usually analyze separately.

1. **Evaluation feedback.** The selector needs a concentration model for noisy estimates of current downstream performance. The bound should remain valid when contexts are chosen adaptively.
2. **Training response.** Training on one context can change performance on other contexts. A theorem needs a transfer model, such as Lipschitz similarity, a shared representation, or a bounded cross-context influence operator.
3. **Distribution-policy complexity.** The result should depend on a meaningful measure of the restricted class: parametric dimension, covering number, policy-class complexity, Wasserstein diameter, or occupancy-polytope geometry.
4. **Cost accounting.** Evaluation samples, training samples, and environment interactions should be separated. A selector may have low regret but spend most of its budget estimating candidate performance.

One possible target is a bound of the form

$$\text{Reg}_T(\Phi) \lesssim \text{complexity}(\Phi)\sqrt{T} + \text{evaluation error} + \text{transfer error} + \text{drift error},$$

where the terms are defined for a particular model rather than assumed abstractly. The existing literature suggests several ways to instantiate the restricted class:

- normalized neural weighting networks [Gra23];
- Gaussian or narrow parametric context distributions [Kli21, Sat25b];
- particle distributions with Wasserstein trust regions [Kli22b, Kli23];
- teacher policies and replay-buffer selectors [Den20, Jia21, Par22];
- flow-feasible occupancy distributions [Hu26].

The main unresolved issue is that the learner’s loss is not a fixed per-round loss. It is the result of a changing policy trained on earlier selected data. Standard online convex optimization therefore cannot be applied without a model of delayed, transferable, and possibly negative training effects.

Conclusions

The search supports a focused conclusion rather than a single canonical algorithm.

- Optimal transport gives a principled way to restrict and move training distributions, with useful transfer and target-alignment results, but not cumulative regret [Hua22b, Kli22b, Kli23].

- Adversarial and replay-based curricula give the clearest regret language, but their guarantees are equilibrium statements rather than finite-time bounds for a restricted shaping class [Den20, Jia21, Par22].
- Bilevel data-selection methods most directly optimize downstream performance, yet their practical neural algorithms lack formal online guarantees [Gra23].
- Exploration and data-acquisition theory supplies the strongest PAC and allocation results, but under objectives and structural assumptions that differ from target-distribution policy improvement [Tar21, Hu26].

The open problem is to combine these strands: an online learner should choose a distribution from a restricted class, observe noisy downstream policy performance, account for transfer across contexts, and achieve a regret bound against the best shaper in that class. The current literature provides mechanisms for each component, but no result that analyzes the complete problem.

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